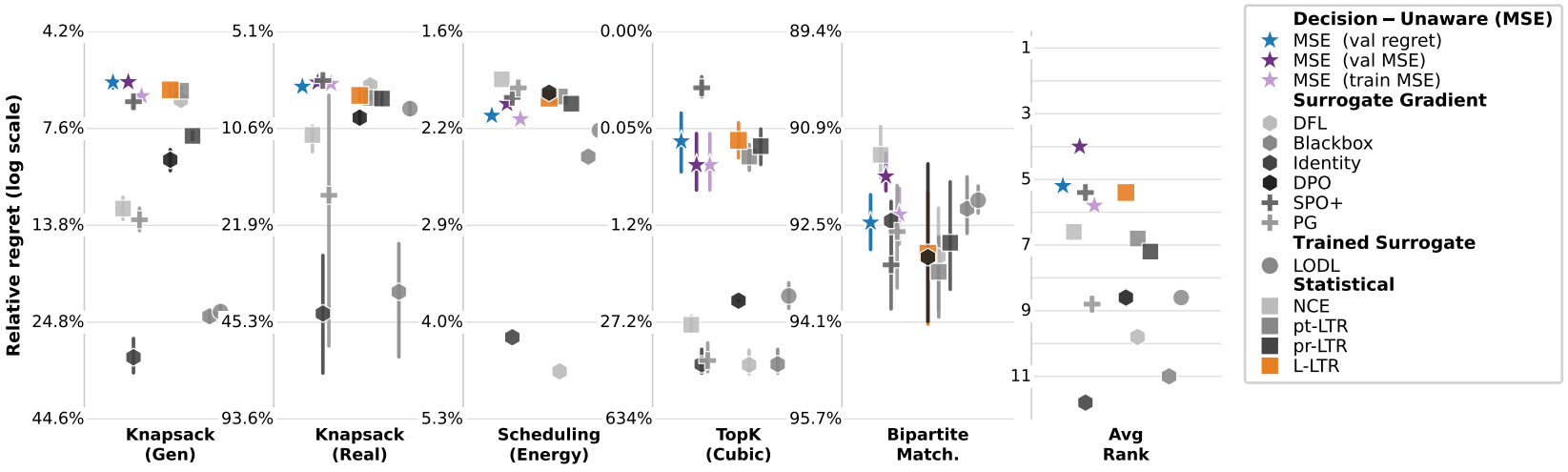
Benchmark re-run — Classic tasks — mean ± 95% bootstrap CI of mean — Relative regret % (best ↑)
Relative regret (log scale)
4.2%
7.6%
13.8%
24.8%
44.6%
5.1%
10.6%
21.9%
45.3%
93.6%
1.6%
2.2%
2.9%
4.0%
5.3%
0.00%
0.05%
1.2%
27.2%
634%
89.4%
90.9%
92.5%
94.1%
95.7%
1
3
5
7
9
11
Knapsack (Gen)
Knapsack (Real)
Scheduling (Energy)
TopK (Cubic)
Bipartite Match.
Avg Rank
Decision – Unaware (MSE)
MSE (val regret)
MSE (val MSE)
MSE (train MSE)
Surrogate Gradient
DFL
Blackbox
Identity
DPO
SPO+
PG
Trained Surrogate
LODL
Statistical
NCE
pt-LTR
pr-LTR
L-LTR